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Topic 1: Key Concepts in Biology

Edexcel GCSE (9-1) Biology (1BI0) — Paper 1

1.1 Microscopes and Magnification

> A **microscope** is an instrument used to view objects that are too small to be seen with the naked eye. It works by using lenses (in a light microscope) or electron beams (in an electron microscope) to produce a magnified image of a specimen.

The development of microscopes transformed biology. In the 1600s, Robert Hooke used an early light microscope to observe cork cells — the first time anyone had seen cells. Antonie van Leeuwenhoek improved lenses and observed bacteria ("animalcules") for the first time. As microscope technology improved, scientists were able to see finer and finer detail: first cells, then organelles, then the internal structure of organelles themselves. The key idea is that as **resolution** improves, we can see more detail.

Magnification and Resolution

Term	Definition	Key point
Magnification	How many times larger an image appears compared to the actual size of the specimen	A higher magnification makes the image bigger but does not necessarily make it clearer.
Resolution	The shortest distance between two points that can still be distinguished as separate	Higher resolution = more detail visible. Limited by the wavelength of the radiation used (light vs. electrons).

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Increasing magnification without improving resolution just produces a bigger blurry image. Electron microscopes have much higher resolution than light microscopes because electrons have a much shorter wavelength than visible light, allowing them to resolve much smaller structures.

Light Microscope vs Electron Microscope

Feature	Light Microscope	Electron Microscope
Source	Visible light	Beam of electrons
Maximum magnification	~ ×2000	~ ×2,000,000 (TEM)
Maximum resolution	~ 200 nm (0.2 μm)	~ 0.2 nm
Specimen	Living or dead; can be in colour	Dead only (vacuum required); image is black & white
Cost & size	Cheap, portable, simple	Very expensive, large, complex
Can see	Whole cells, nucleus, chloroplasts, larger organelles	Ribosomes, internal structure of mitochondria, viruses

Exam Tip: A common exam question: "Why can electron microscopes see smaller structures than light microscopes?" Answer: electrons have a **much shorter wavelength** than light, giving a **higher resolution**, so smaller structures can be resolved as separate.

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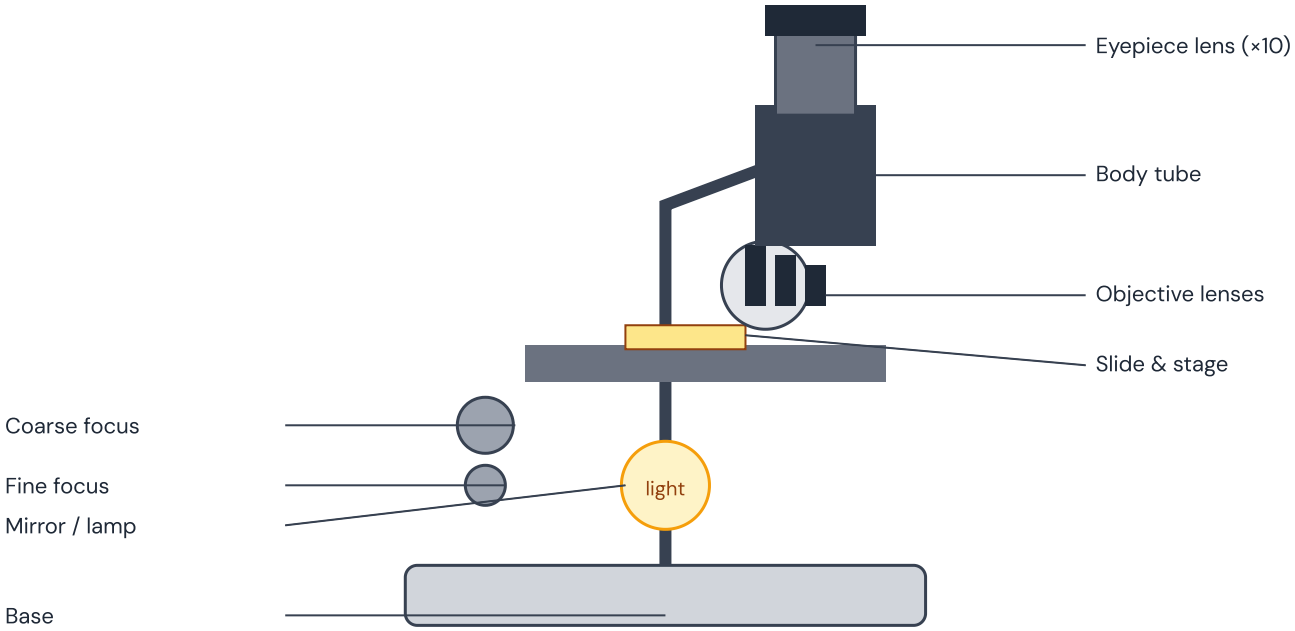
Labelled Light Microscope



Units of Measurement in Biology

Biologists work at a huge range of scales, so you must be comfortable converting between millimetres (mm), micrometres (µm) and nanometres (nm).

Unit	Symbol	Relation to metre	Typical use
millimetre	mm	1×10^{-3} m	Small organisms, ruler measurements
micrometre	µm	1×10^{-6} m	Cells (most 10–100 µm), organelles
nanometre	nm	1×10^{-9} m	Ribosomes, viruses, membranes

Conversions: `1 mm = 1000 µm`, `1 µm = 1000 nm`, `1 mm = 1,000,000 nm`.

The Magnification Equation

Magnification = Image size ÷ Actual size

Rearranged: Image size = magnification × actual size, and Actual size = image size ÷

magnification.

Always ensure image size and actual size are in the **same units** before dividing. Convert to micrometres if the question gives mixed units.

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Worked example 1
  A plant cell appears 40 mm wide in an image taken at ×400 magnification.
  What is the actual width of the cell in µm?

  Image size = 40 mm = 40,000 µm
  Actual = image / magnification = 40,000 / 400 = 100 µm

Worked example 2
  A mitochondrion is 2 µm long. How long does it appear on an
  image taken at ×50,000?

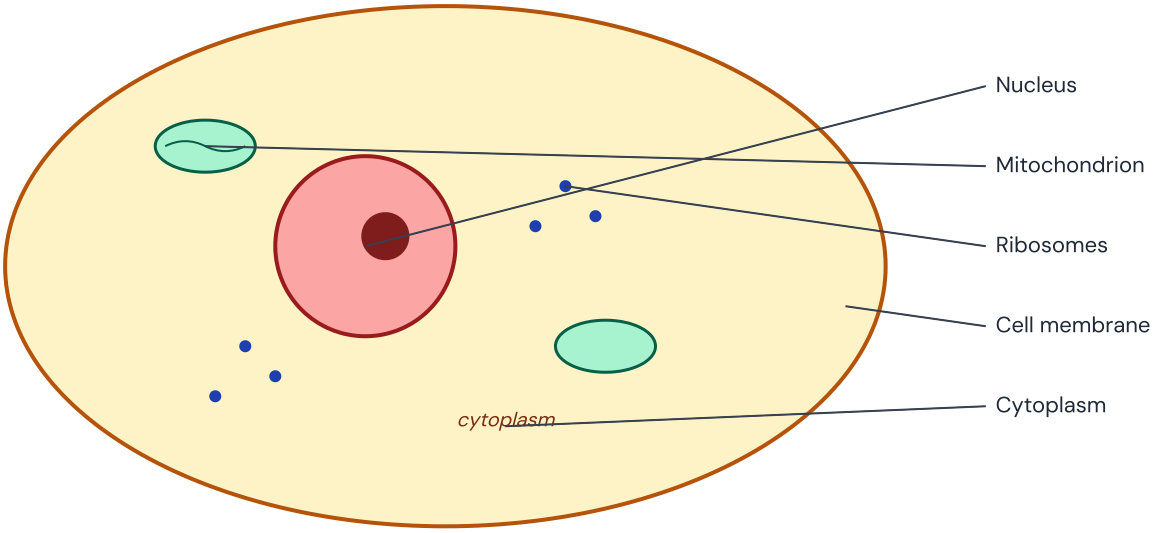
  Image = magnification × actual = 50,000 × 2 = 100,000 µm = 100 mm
```

Exam Tip: Always check units! A wrong answer by a factor of 1000 is almost always a unit conversion error (mm vs µm). Show your working clearly — the final answer is usually only worth 1 mark, but correctly rearranging the equation earns additional marks.

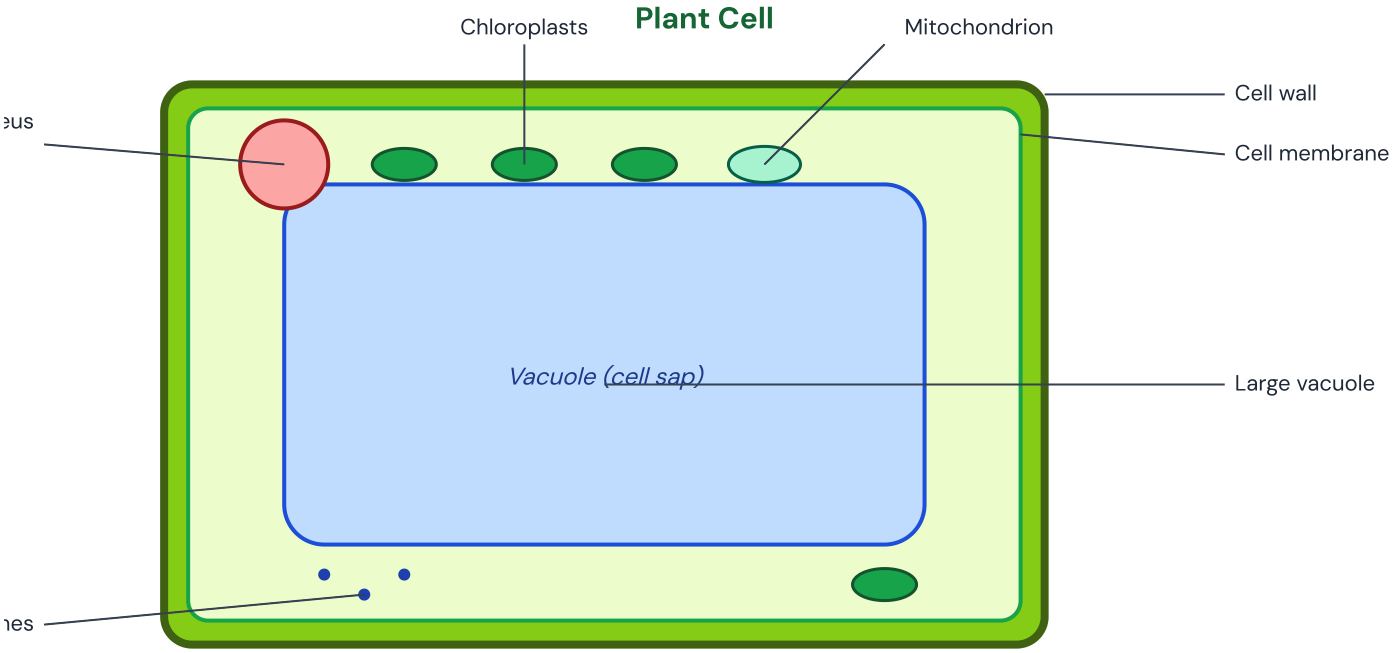
1.2 Animal, Plant and Bacterial Cells

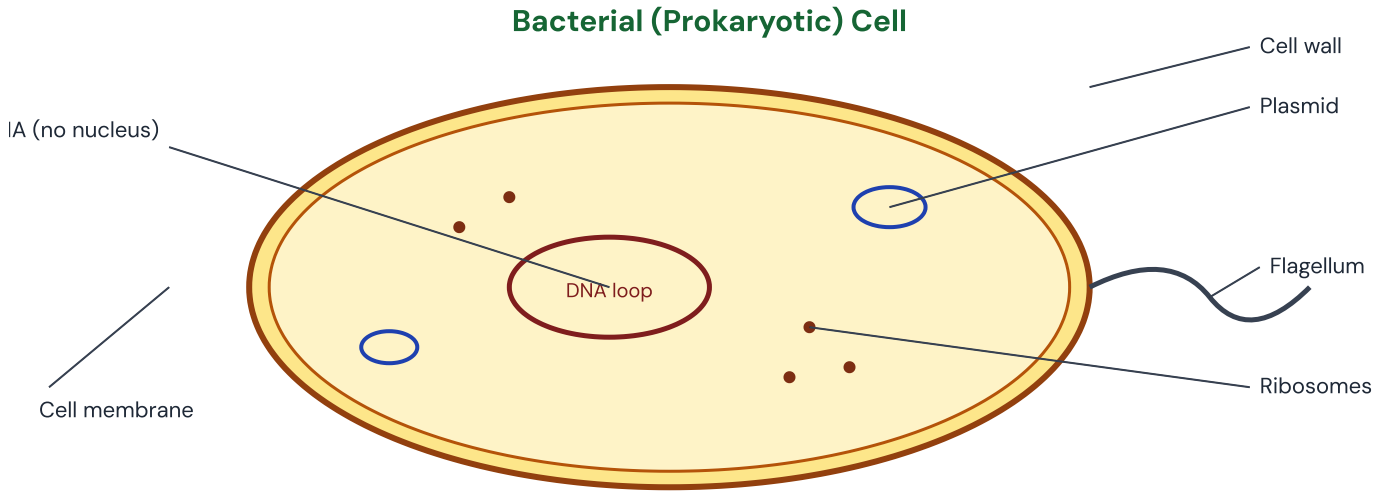
All living organisms are made of cells. There are two main types: **eukaryotic** cells (animal, plant, fungi, protists) which have a true nucleus and membrane-bound organelles, and **prokaryotic** cells (bacteria) which do not.

Animal Cell



Plant Cell





1.3 Eukaryotic vs Prokaryotic Cells

Feature	Eukaryotic (animal / plant)	Prokaryotic (bacteria)
Size	Large (10–100 μm)	Small (1–10 μm)
Nucleus	True nucleus surrounded by membrane	No nucleus — DNA free in cytoplasm
DNA	Linear chromosomes inside nucleus	Single circular DNA loop + plasmids
Membrane-bound organelles	Yes (mitochondria, chloroplasts etc.)	No
Ribosomes	Larger (80S)	Smaller (70S)
Cell wall	Plants only — made of cellulose	Always present — made of peptidoglycan (murein)
Examples	Human cells, plant cells, yeast	E. coli, Streptococcus, Lactobacillus

Subcellular Structures and Their Functions

Organelle	Found in	Function
Nucleus	All eukaryotes	Contains DNA/chromosomes; controls cell activities by coding for proteins
Cytoplasm	All cells	Jelly-like substance where most chemical reactions take place; contains dissolved enzymes
Cell membrane	All cells	Controls what enters and leaves the cell (partially permeable)
Mitochondria	Animal & plant cells	Site of aerobic respiration — releases energy from glucose
Ribosomes	All cells	Site of protein synthesis
Chloroplasts	Plant cells only	Contain chlorophyll; site of photosynthesis
Cell wall	Plants (cellulose) and bacteria (peptidoglycan)	Provides strength & support; prevents cell bursting when turgid
Permanent vacuole	Plant cells	Contains cell sap; maintains turgor pressure to keep cell rigid
Plasmids	Bacteria only	Small rings of DNA carrying extra genes (e.g. antibiotic resistance)
Flagellum	Some bacteria & sperm cells	Long whip-like structure used for movement

Exam Tip: "What do animal and plant cells have in common?" Answer: nucleus, cell membrane, cytoplasm, mitochondria, ribosomes. Plant cells **also** have cell wall, chloroplasts, and a permanent vacuole. Don't say "plant cells have more organelles" — be specific.

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1.4 Specialised Cells

Multicellular organisms contain many types of cell. Each type is **specialised** — its structure is adapted to carry out a particular function efficiently. This is called the **structure-function relationship**.

Sperm Cell

- **Function:** male gamete — carries genetic material to fertilise an egg.
- **Long tail (flagellum):** allows the sperm to swim towards the egg.
- **Many mitochondria** in the middle section: release energy from respiration for tail movement.
- **Acrosome** (at the head): contains digestive enzymes that break through the egg's outer membrane.
- **Haploid nucleus**: carries 23 chromosomes so that fertilisation restores the full 46.
- **Streamlined head**: reduces resistance while swimming.

Egg Cell (Ovum)

- **Function:** female gamete — carries genetic material and provides nutrients for the early embryo.
- **Large cytoplasm** full of nutrients (yolk): supplies the embryo with energy before implantation.
- **Jelly-like outer coat (zona pellucida)**: changes chemically after fertilisation so that only one sperm can enter — prevents polyspermy.
- **Haploid nucleus**: contains 23 chromosomes.

Ciliated Epithelial Cell

- **Function:** lines surfaces such as the trachea, bronchi, and fallopian tubes.
- **Cilia** on the surface: tiny hair-like projections that beat in a rhythmic wave to move mucus (with trapped dust/bacteria) out of the lungs, or an egg down the fallopian tube.
- **Many mitochondria**: provide energy for the continuous beating of the cilia.

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Exam Tip: For specialised cell questions, always link **structure to function**. Don't just say "sperm has a tail"; say "the tail allows the sperm to swim to the egg." Don't just list features — **explain the adaptation**.

1.5 Enzymes — Lock-and-Key Model

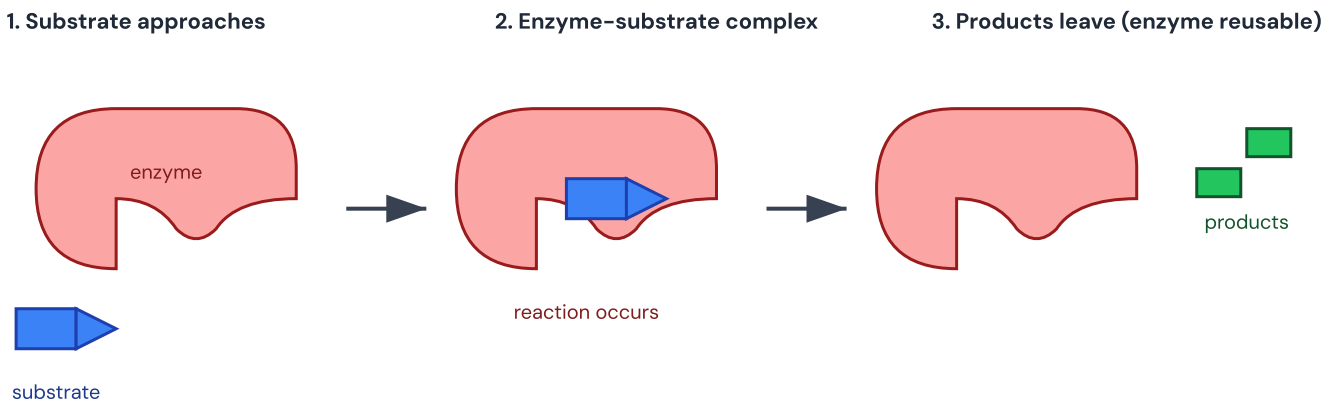
An **enzyme** is a biological catalyst — a protein that speeds up a chemical reaction without being used up. Enzymes are essential for life because reactions in cells would otherwise be far too slow at body temperature.

Every enzyme has a region called the **active site**. Only molecules with a matching shape (the **substrate**) can fit into the active site. This is called the **lock-and-key model**: the enzyme is the lock and the substrate is the key — only the correct key fits.

1. Substrate collides with the enzyme and fits into the active site, forming an **enzyme–substrate complex**.
2. The reaction occurs — the substrate is broken down (or built up) into product(s).
3. Product leaves the active site. The enzyme is unchanged and can be reused.

Because the shape of the active site is so specific, each enzyme catalyses only one reaction — this is called **enzyme specificity**.

Lock-and-Key Model of Enzyme Action



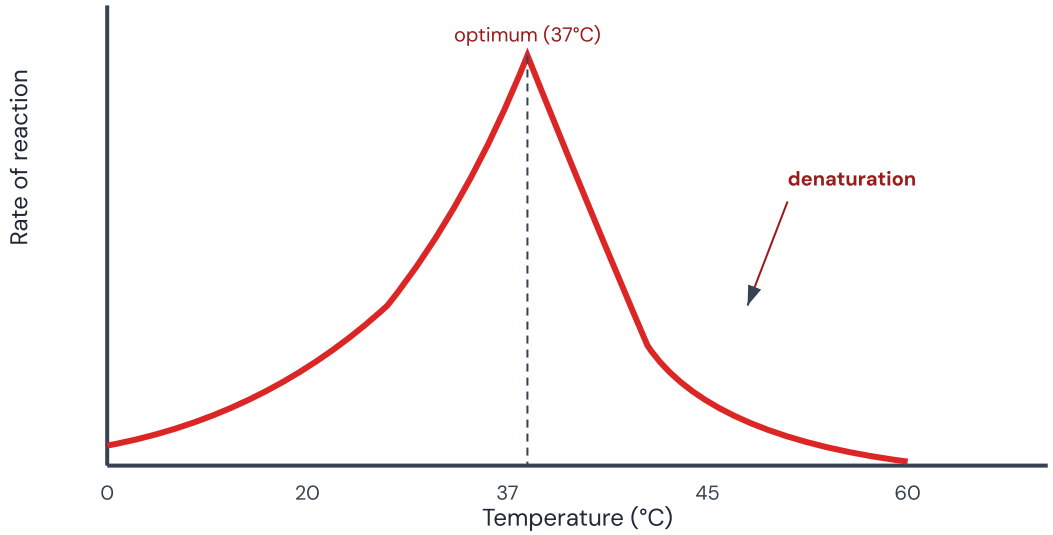
1.6 Factors Affecting Enzyme Activity

Three main factors affect how fast an enzyme catalyses a reaction: **temperature, pH,** and **substrate concentration**.

Effect of Temperature

- At **low temperatures**: molecules have little kinetic energy, so fewer successful collisions — rate is low.
- As temperature rises: molecules move faster, more collisions per second — rate increases.
- At the **optimum temperature** (around 37°C for human enzymes): rate is at its maximum.
- Above the optimum: bonds holding the enzyme's 3D shape break. The active site changes shape so the substrate no longer fits — the enzyme is **denatured**. Denaturation is **permanent**.

Rate of Reaction vs Temperature

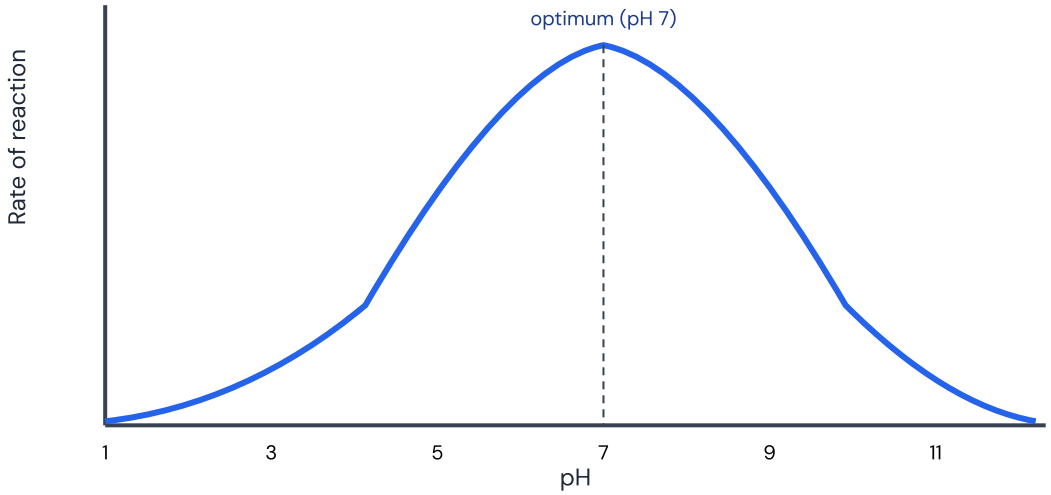


Effect of pH

- Each enzyme has an **optimum pH**. Most human enzymes work best around pH 7, but **pepsin** (in the stomach) works best at pH 2, and enzymes in the small intestine work best around pH 8.

- Above or below the optimum pH, the H^+ and OH^- ions disrupt bonds holding the enzyme together, changing the active site's shape and denaturing the enzyme.

Rate of Reaction vs pH (amylase, pH 7 optimum)



Effect of Substrate Concentration

- As substrate concentration increases, the rate of reaction increases because more collisions between substrate and enzyme occur per second.
- At high substrate concentration, all enzyme active sites are occupied — the enzyme is the limiting factor. Adding more substrate has no effect; the graph **levels off (plateaus)**.
- Adding more enzyme at this point would increase the rate again.

> **Exam Tip:** Never say an enzyme is "killed" by high temperature — enzymes aren't alive! The correct term is **denatured**: the 3D shape is permanently changed so the substrate no longer fits. Denaturation is irreversible.

1.7 Enzymes in Digestion

Enzyme class	Example	Substrate	Products	Site of action

Carbohydrase	Amylase	Starch	Maltose → glucose	Salivary glands, pancreas, small intestine
Protease	Pepsin	Protein	Amino acids	Stomach (pH 2)
Protease	Trypsin	Protein	Amino acids	Pancreas / small intestine (pH 8)
Lipase	Lipase	Lipid (fat/oil)	Fatty acids + glycerol	Pancreas, small intestine

Bile (produced in the liver, stored in the gall bladder) is not an enzyme. It **emulsifies** lipids — breaks them into small droplets, increasing the **surface area** for lipase to act on. Bile is also alkaline, which **neutralises** stomach acid so small intestine enzymes can work at their optimum pH.

1.8 Calculating Rate of Reaction

Rate of reaction = amount of substrate used (or product formed) ÷ time taken

Units depend on the measurement used — cm^3/s, g/min, mol/s, etc.

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Worked example
  In an experiment, 36 cm³ of oxygen is produced by catalase
  in 60 seconds. Calculate the rate.

  Rate = 36 / 60 = 0.6 cm³/s

Worked example 2
  A starch solution is completely digested by amylase in 4 minutes.
  The starting amount was 20 mg.

  Rate = 20 / 4 = 5 mg/min
```

1.9 Food Tests — Biological Molecules

Molecule tested	Test	Method	Positive result
Starch	Iodine test	Add a few drops of yellow/orange iodine solution to the sample	Blue-black colour
Reducing sugars (e.g. glucose, maltose)	Benedict's test	Add Benedict's solution (blue); heat in water bath at 75°C for ~5 minutes	Brick-red (or orange/yellow/green) precipitate
Protein	Biuret test	Add a few drops of Biuret solution (blue, copper sulfate + sodium hydroxide) to the sample	Purple / lilac colour
Lipids	Ethanol emulsion test	Dissolve the sample in ethanol, then add water and shake	Cloudy white emulsion

Exam Tip: Benedict's test is **semi-quantitative**: the colour after heating indicates the approximate amount of reducing sugar (blue → green → yellow → orange → brick-red as concentration increases). You do **not** need to heat Biuret, iodine, or the emulsion test — only Benedict's.

1.10 Uses of Energy in the Body

Energy released from glucose during respiration is used for many processes:

- **Metabolism** — building up larger molecules (e.g. proteins from amino acids, starch from glucose).

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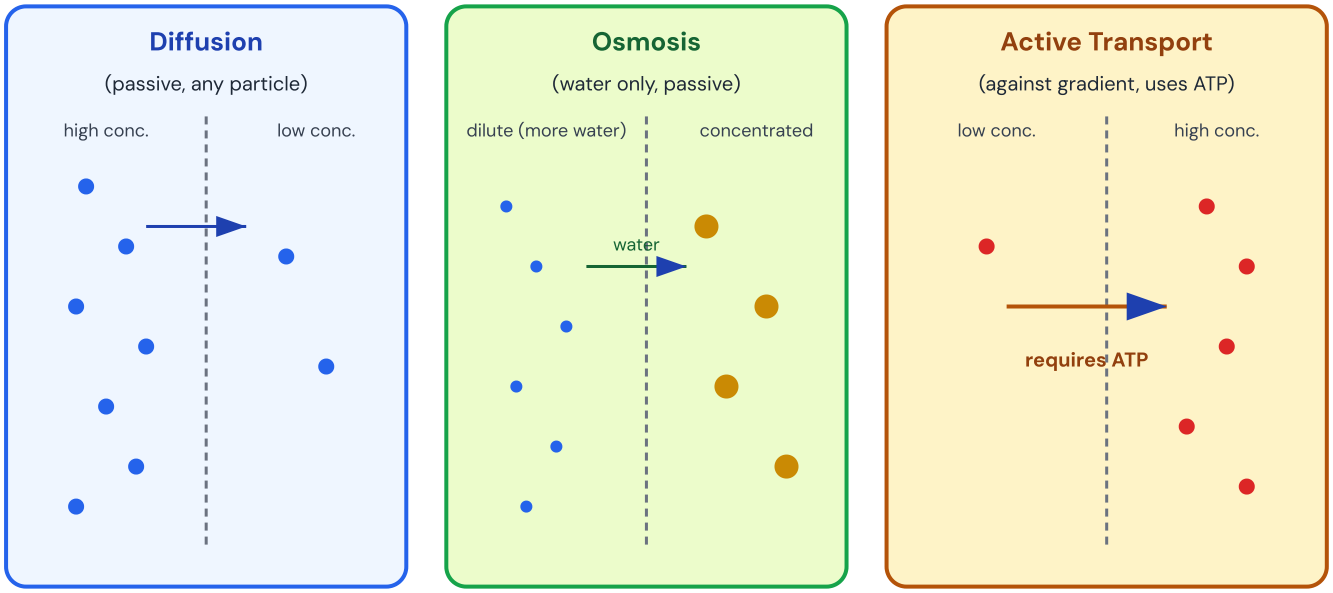
- **Movement** — muscle contraction requires ATP (energy).
- **Keeping warm** — maintaining body temperature at 37°C (especially in mammals and birds).
- **Making molecules** — synthesising enzymes, hormones, DNA, and other biological molecules.
- **Active transport** — moving substances against a concentration gradient across membranes.
- **Cell division** — growth, repair, and reproduction.

1.11 Transport Across Cell Membranes

Cells must constantly exchange substances with their surroundings — taking in oxygen, glucose, water and ions, and getting rid of carbon dioxide and urea. There are three main transport processes:

Diffusion vs Osmosis vs Active Transport



Diffusion

Diffusion is the **net movement** of particles from a region of **high concentration to low concentration**, down the concentration gradient. It is a **passive** process — no energy from respiration is required.

Examples in biology: oxygen diffusing from alveoli into blood; carbon dioxide diffusing from cells into blood; urea diffusing from cells into blood.

Factors Affecting Rate of Diffusion (Fick's Law)

Rate of diffusion ∝ (surface area × concentration difference) ÷ distance

- **Surface area**: larger surface area = more particles can move across at once = faster rate (e.g. alveoli are folded into tiny sacs to increase SA).
- **Concentration difference (gradient)**: steeper gradient = greater net movement = faster rate.
- **Distance (membrane thickness)**: shorter distance = faster rate (e.g. alveolus walls are just one cell thick).
- **Temperature**: higher temperature = particles have more kinetic energy = faster rate.

Osmosis

Osmosis is the movement of **water molecules** from a region of **higher water concentration (dilute solution) to lower water concentration (concentrated solution)**, across a **partially permeable membrane**. It is passive.

Solution	Meaning	Effect on animal cell	Effect on plant cell
Hypotonic	Less solute (more water) than the cell	Water enters; cell swells and may burst (**lysis**)	Water enters vacuole; cell becomes **turgid** (firm) — cell wall prevents bursting
Isotonic	Equal solute/water to the cell	No net movement; cell unchanged	No net movement; cell normal

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Hypertonic	More solute (less water) than the cell	Water leaves; cell shrinks and shrivels (**crenated**)	Water leaves vacuole; cell becomes **plasmolysed** — membrane pulls away from cell wall

Active Transport

> **Active transport** is the movement of particles **against a concentration gradient** (from low to high concentration) across a membrane. It requires **energy from respiration (ATP)**.

Examples:

- **Root hair cells** absorb mineral ions (e.g. nitrate, magnesium) from the soil even when the soil has a lower concentration than the cell.
- Glucose is absorbed from the intestine into blood after the gradient has reversed (once blood glucose is higher than gut glucose).
- Sodium-potassium pumps in nerve cells maintain ion gradients essential for electrical signals.

> **Exam Tip:** The three transport processes differ in three key ways: (1) **What moves** — any particle (diffusion), water only (osmosis), specific ions/molecules (active transport). (2) **Direction** — down the gradient (diffusion, osmosis) vs against it (active transport). (3) **Energy** — passive (diffusion, osmosis) vs requires ATP (active transport). Always state which process is which in exam answers.

Key Definitions — Glossary

Term	Definition
Magnification	Number of times an image is larger than the actual specimen (image size ÷ actual size).

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Resolution	The smallest distance between two points that can still be seen as separate.
Eukaryotic cell	A cell with a true nucleus and membrane-bound organelles (animal, plant, fungi).
Prokaryotic cell	A cell without a true nucleus; DNA is free in the cytoplasm (bacteria).
Enzyme	A protein that speeds up a chemical reaction without being used up (a biological catalyst).
Active site	The region on an enzyme where the substrate binds and the reaction occurs.
Denaturation	Permanent change in the 3D shape of an enzyme's active site so the substrate no longer fits.
Diffusion	Net movement of particles from high to low concentration; passive.
Osmosis	Net movement of water across a partially permeable membrane from dilute to concentrated solution; passive.
Active transport	Movement of particles against a concentration gradient; requires energy (ATP).
Turgid	A plant cell that is firm and swollen with water (in a hypotonic solution).
Plasmolysed	A plant cell that has lost so much water that the membrane pulls away from the cell wall.

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